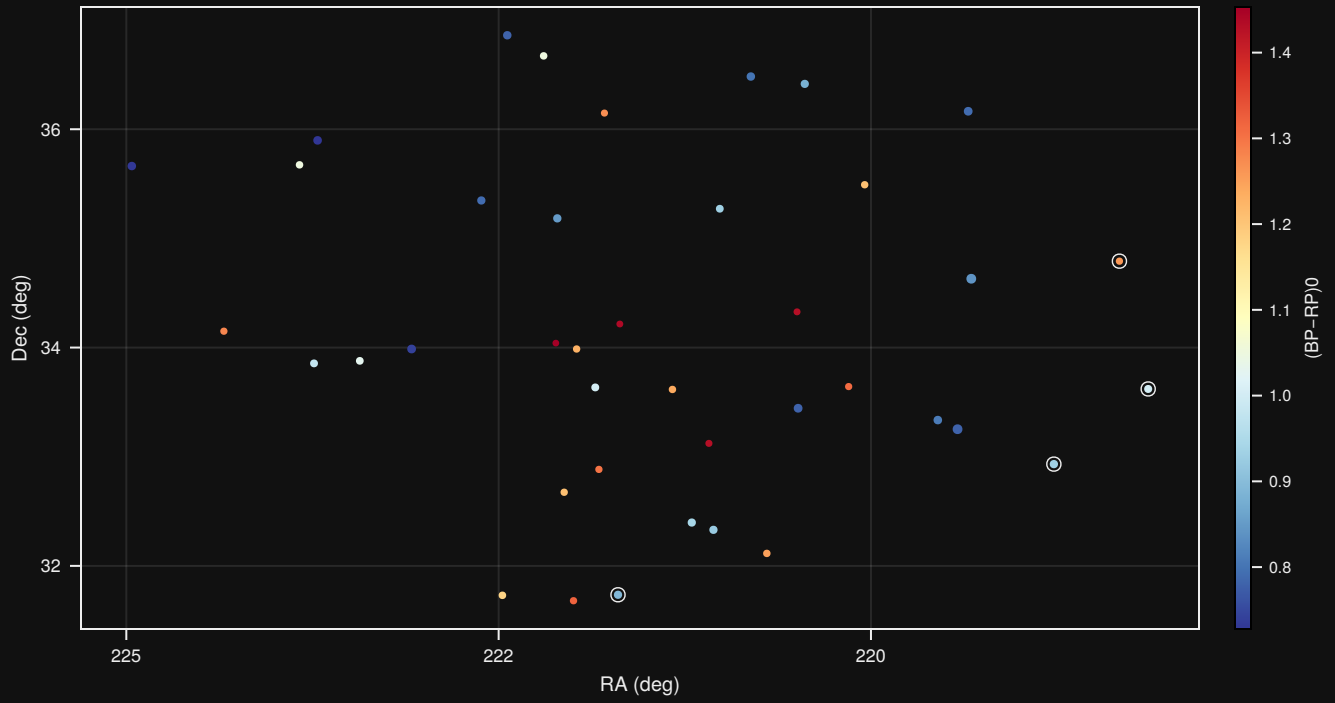
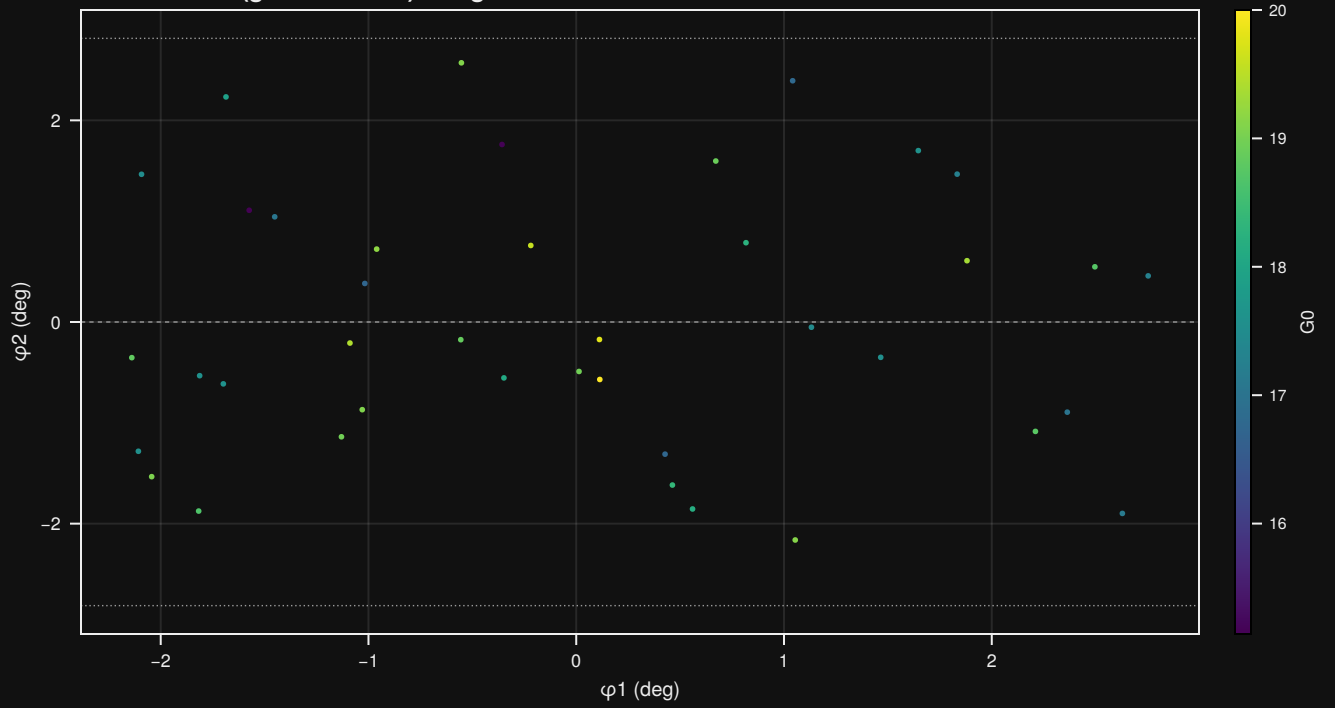
New-19

MEMBERSHIP	
Stars in catalog	40
with Gaia GSP-Phot	29 (72%)
with radial velocity	4
Stream id (sID)	60
GEOMETRY · great-circle fit	
Centroid RA, Dec	221.4, 34.0 deg
Centroid l, b	56.0, 64.4 deg
Pole RA, Dec	116.0, 21.4 deg
Length (φ1, 2–98%)	4.9 deg
Width σ(φ2), robust	1.41 deg
KINEMATICS	
pmRA* med ± σ	-3.21 ± 0.78 mas/yr
pmDE med ± σ	-3.12 ± 1.17 mas/yr
μφ1* med ± σ	-4.28 ± 1.09 mas/yr
μφ2 med ± σ	1.62 ± 1.07 mas/yr
Parallax (cat.) med	0.302 ± 0.105 mas
Parallax (wtd, ZP-corr.)	0.350 ± 0.017 mas
Distance	d = 2.9 ± 0.1 kpc
Physical length	0.24 kpc
RV median ± σ(MAD)	-37.9 ± 41.2 km/s
RV gradient dV/dφ1	—
PHOTOMETRY · POPULATIONS	
G0 range (median)	15.1–20.0 (18.2)
(BP−RP)0 median	1.00
GSP-Phot [M/H] (n=29)	-0.68 ± 0.22
GSP-Phot Teff median	5242 K
RUWE > 1.4	3.4 %
PM clusters (largest)	3 (18 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	40 | 40
n_RV catalog | paper	4 | 4
<[Fe/H]> paper	-0.79
Stellar mass (paper)	100 Msun
Pole offset vs paper	1.6 deg
Remark	Compact, velocities unclear

Sky map · catalog members (n = 40) · size ∝ brightness · ring = has RV

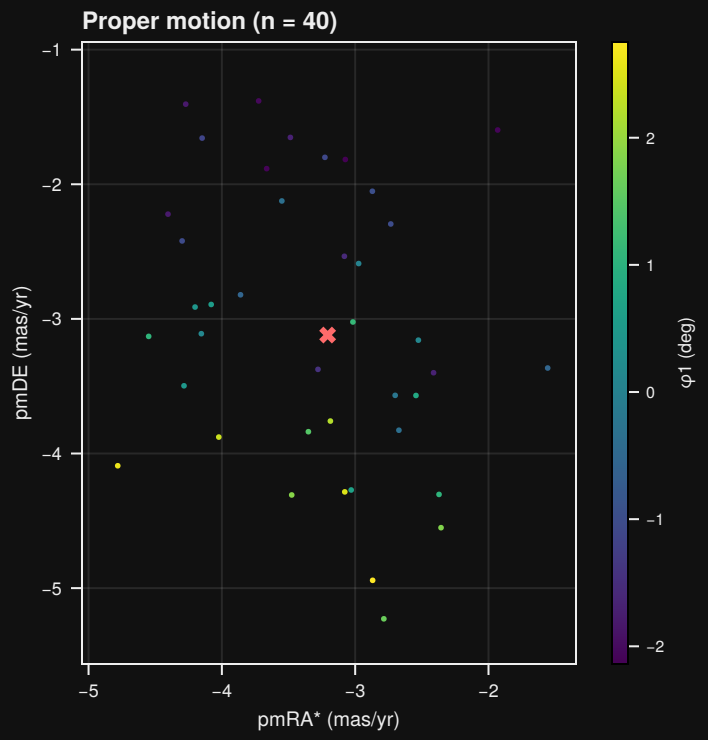


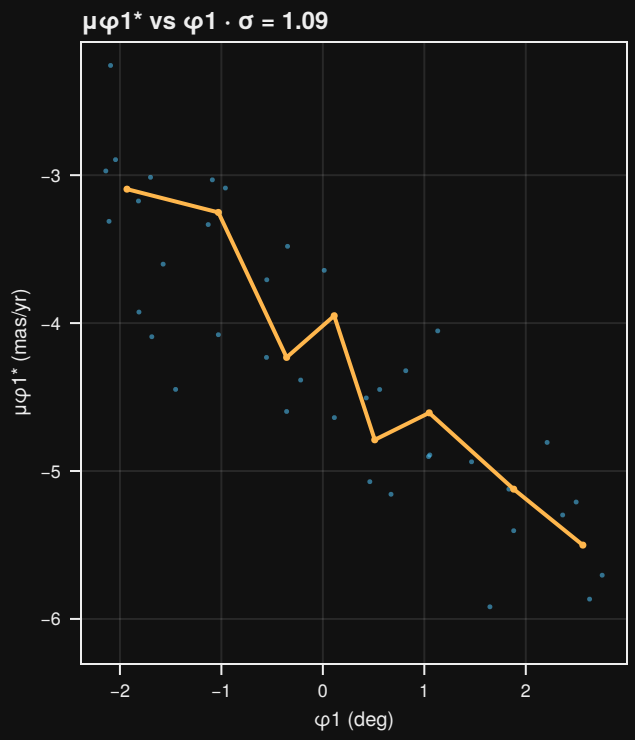
Stream frame (great-circle fit) · length 4.9° · width σ 1.41°

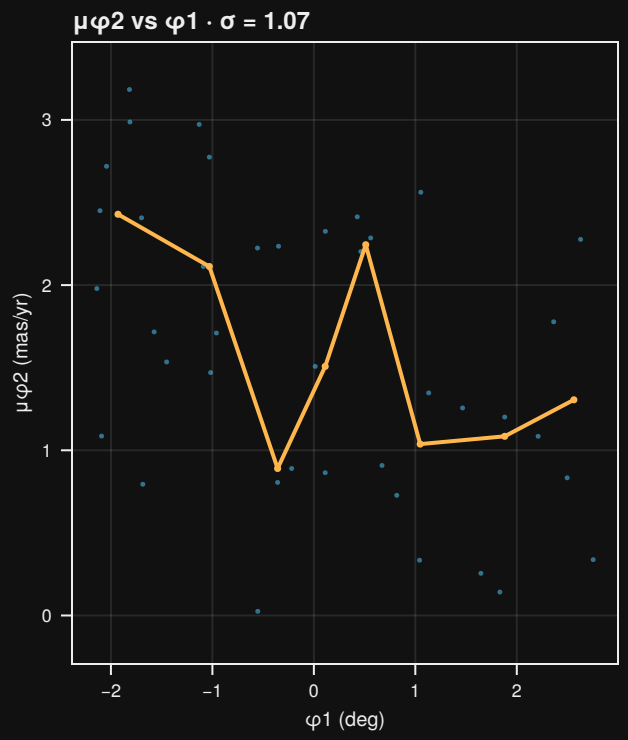


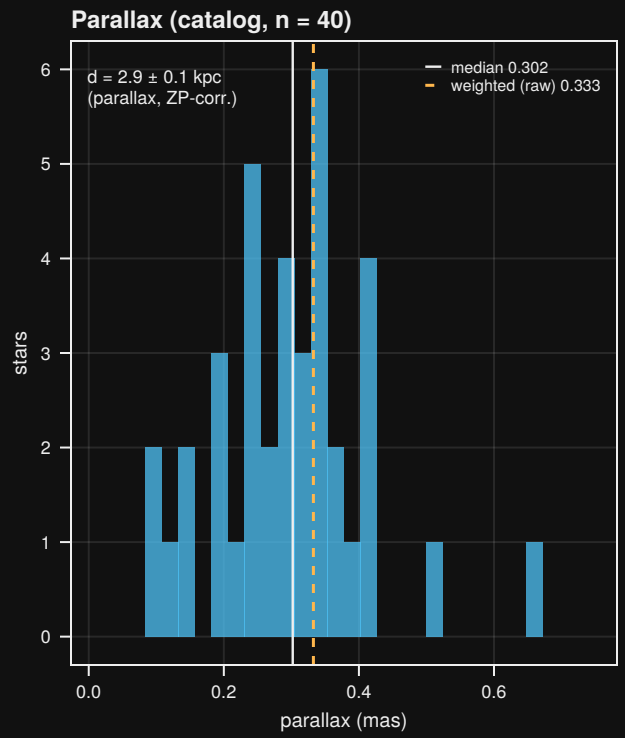
New-19

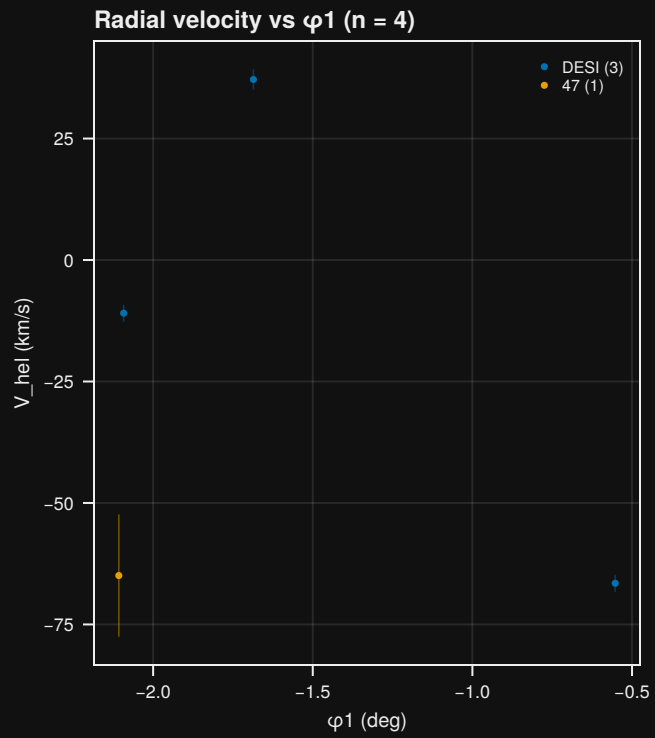
Kinematics · proper motion · parallax · radial velocity

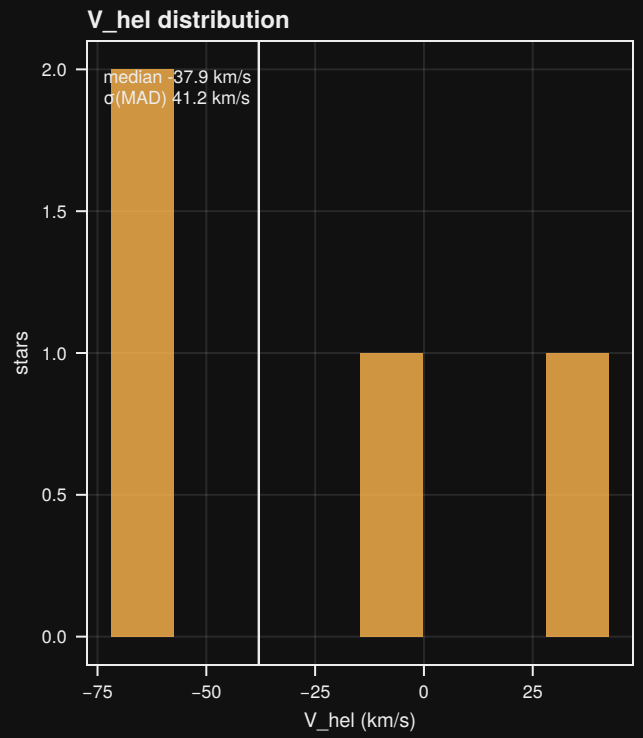






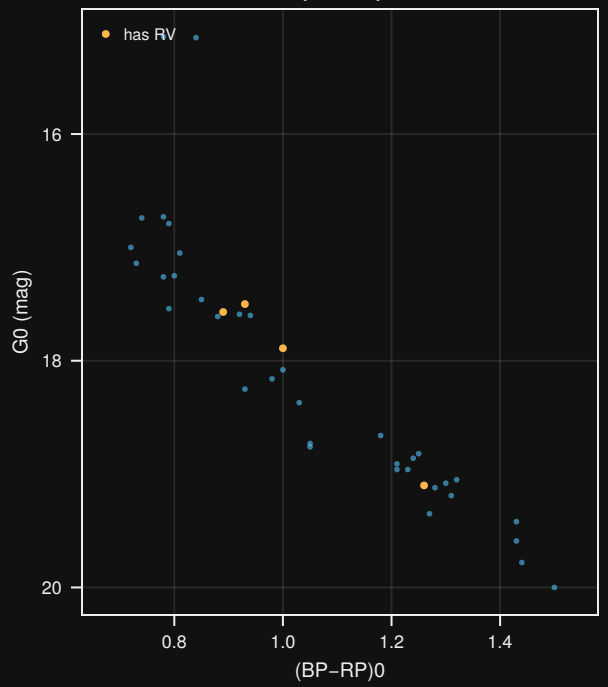




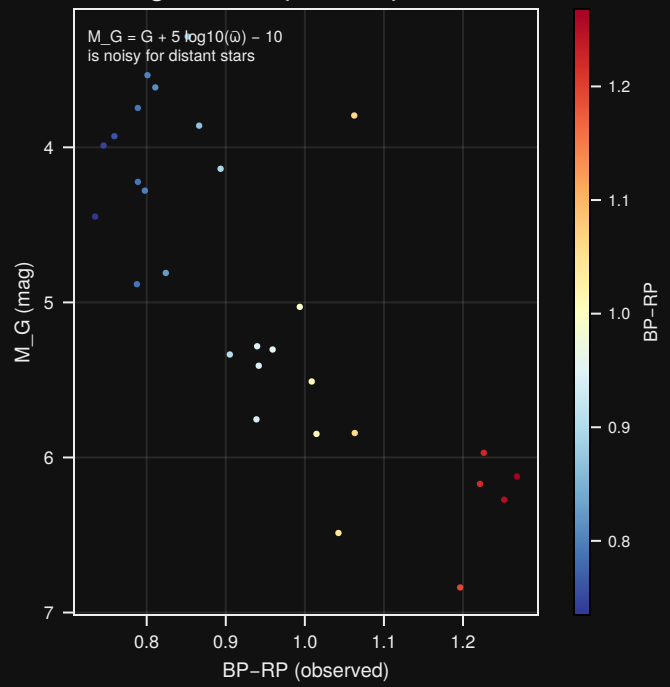


New-19

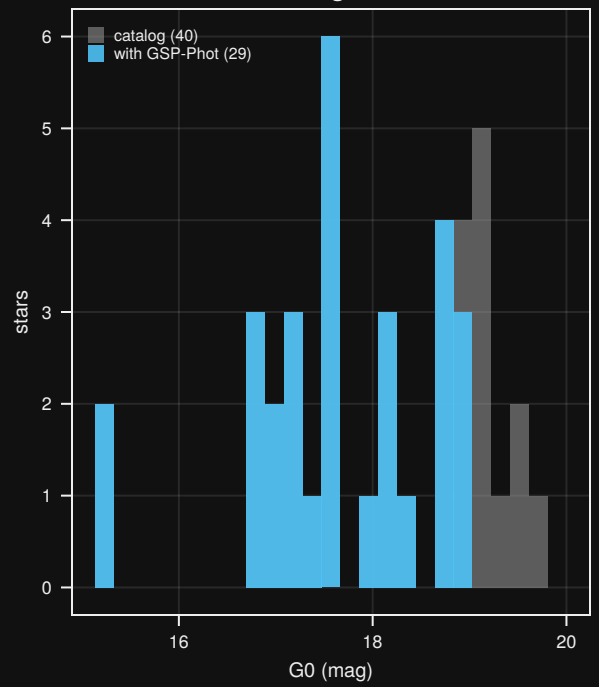
CMD · dereddened (n = 40)



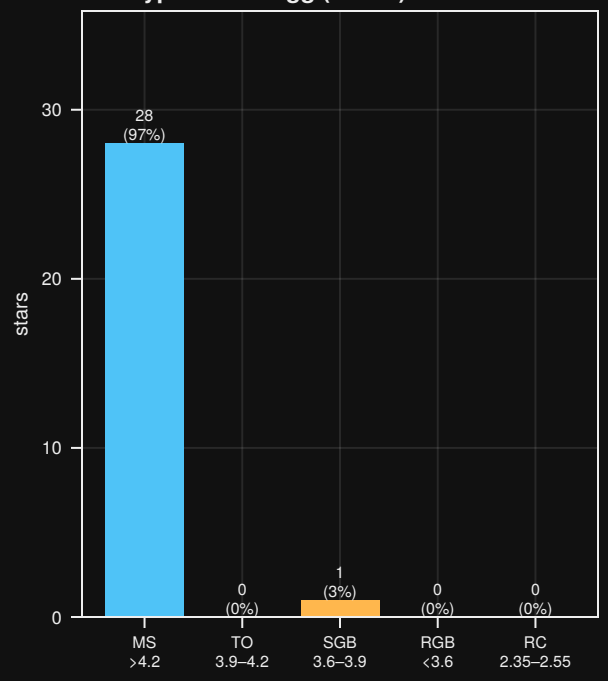
HR diagram · ϖ>0 (n = 29/29)



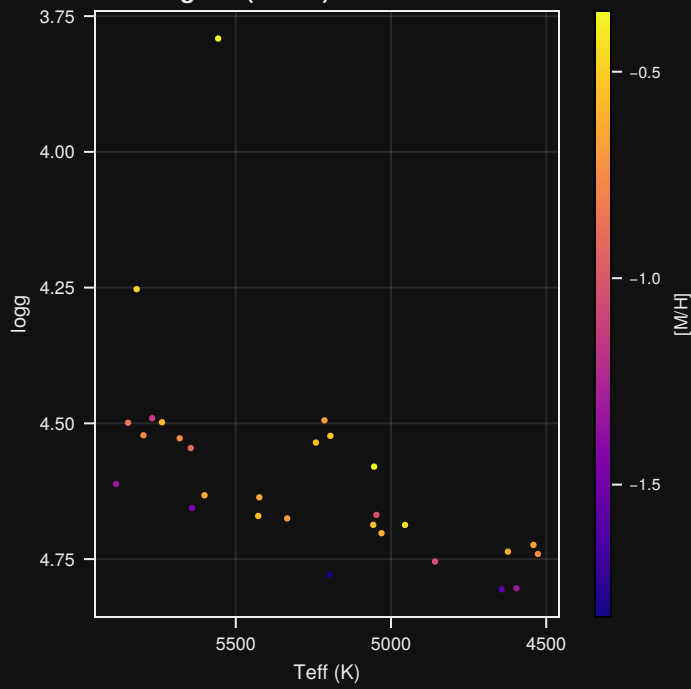
G0 & GSP-Phot coverage



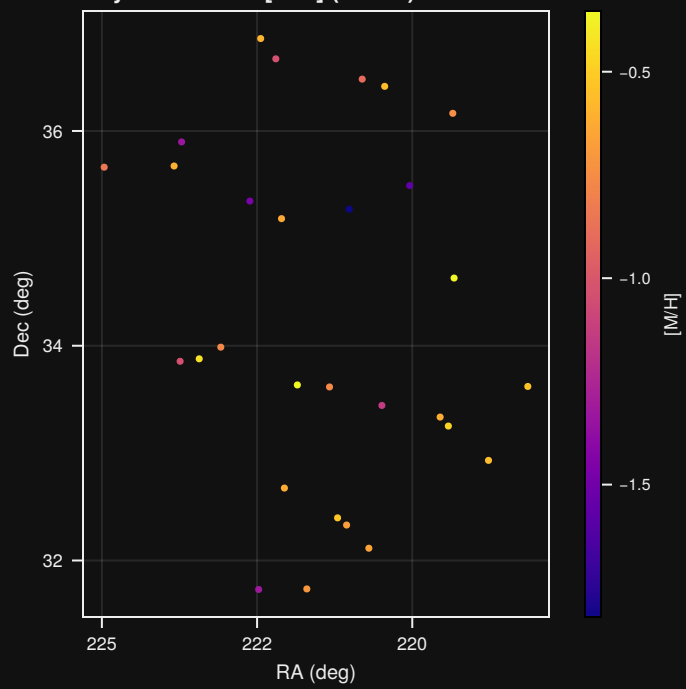
Star types from logg (n = 29)



Kiel diagram (n = 29)



Sky · GSP-Phot [M/H] (n = 29)

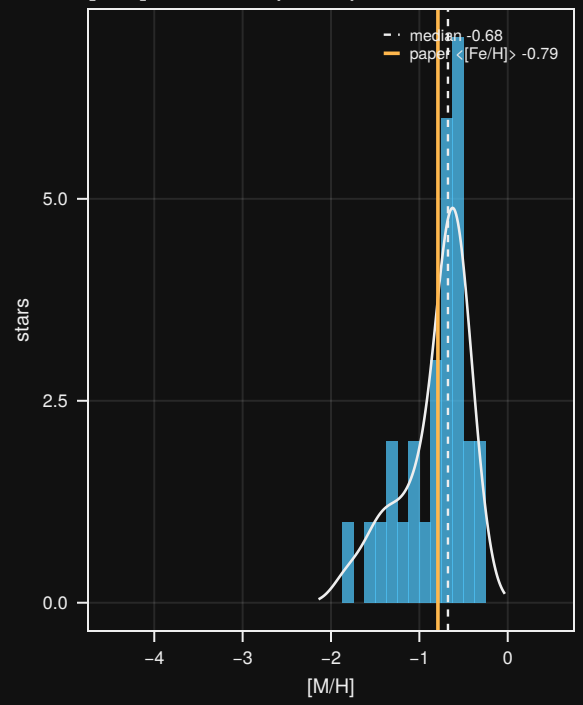


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

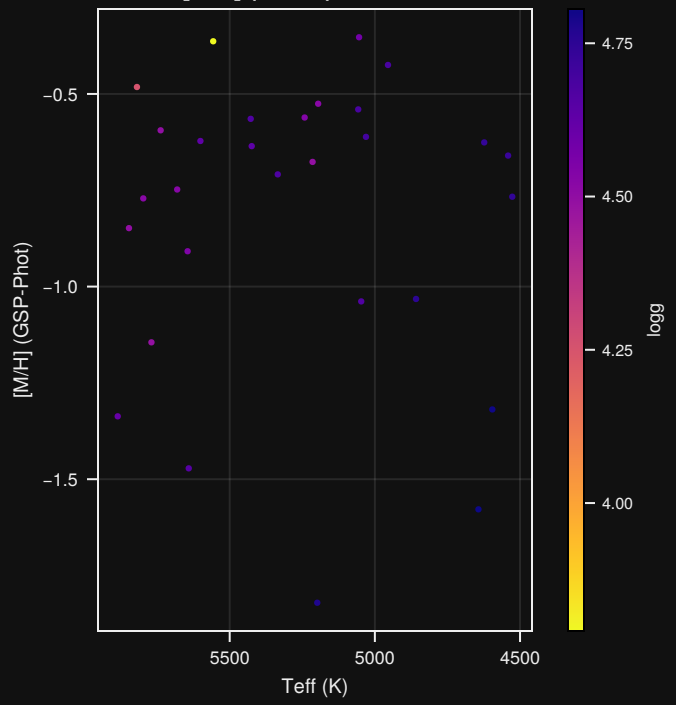
New-19

Metallicity · age · astrometric quality

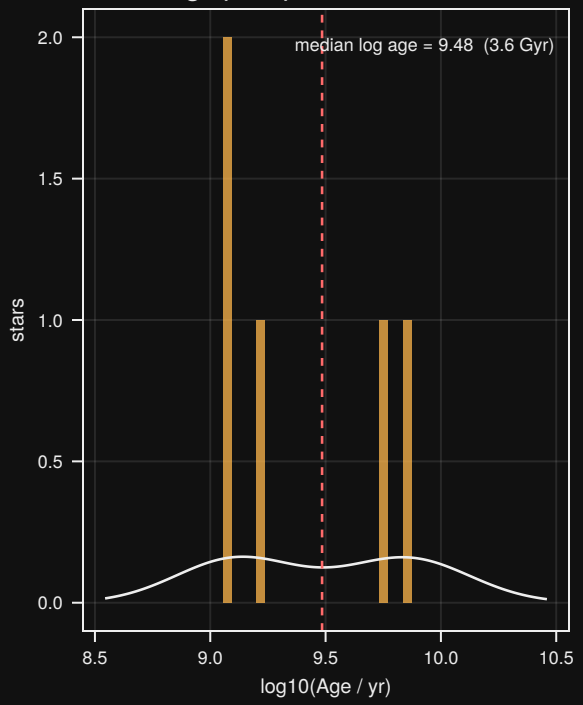
[M/H] GSP-Phot (n = 29)



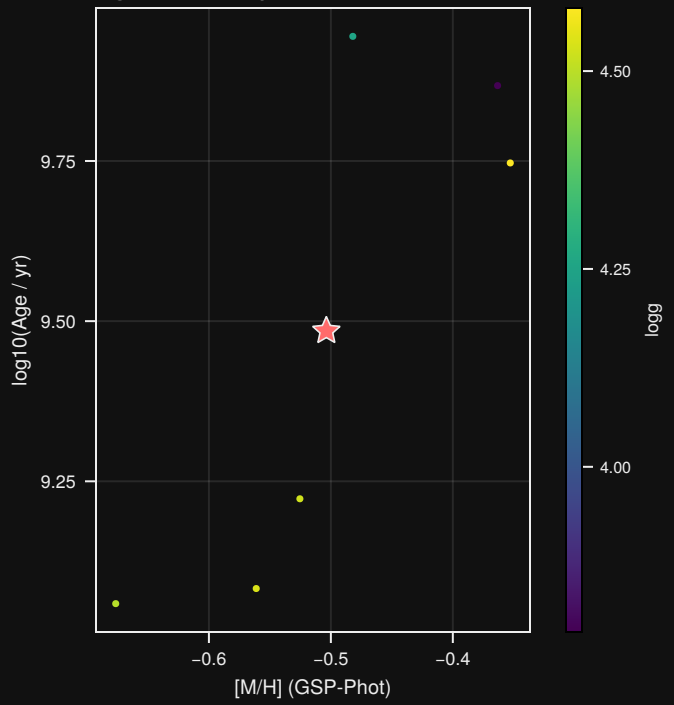
Teff vs [M/H] (n = 29)



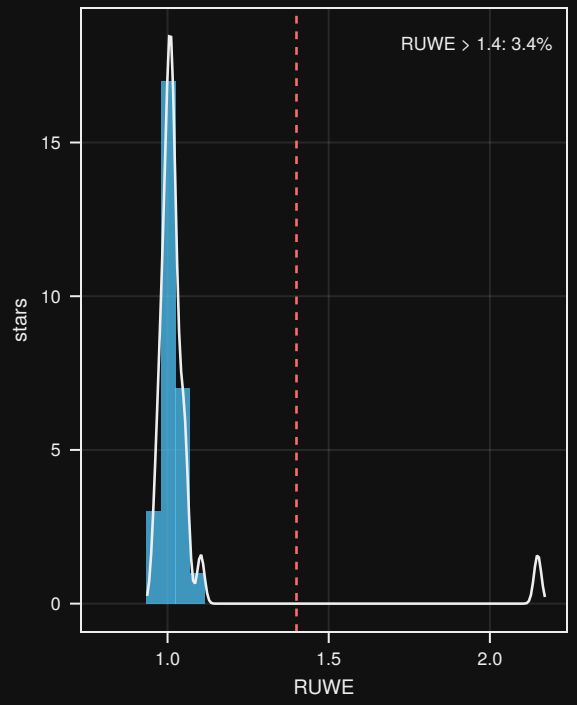
FLAME age (n = 6)



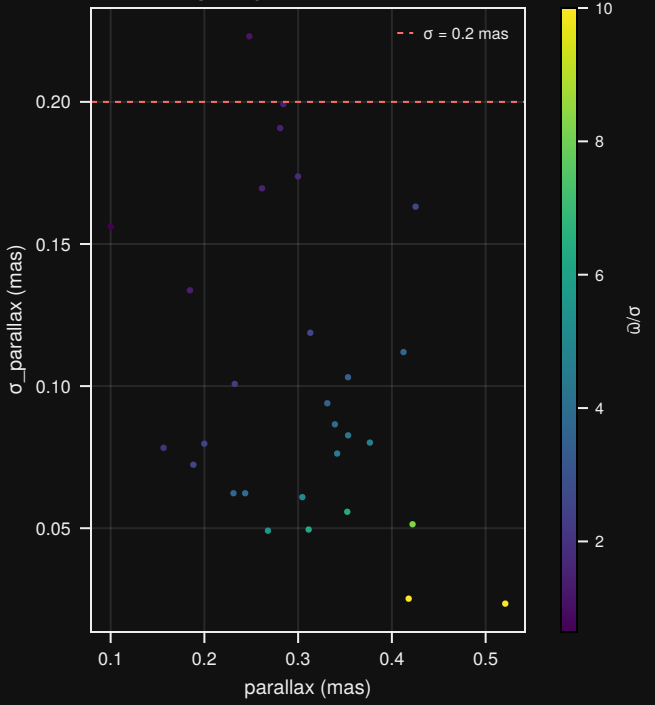
Age–metallicity (n = 6)



RUWE (n = 29)

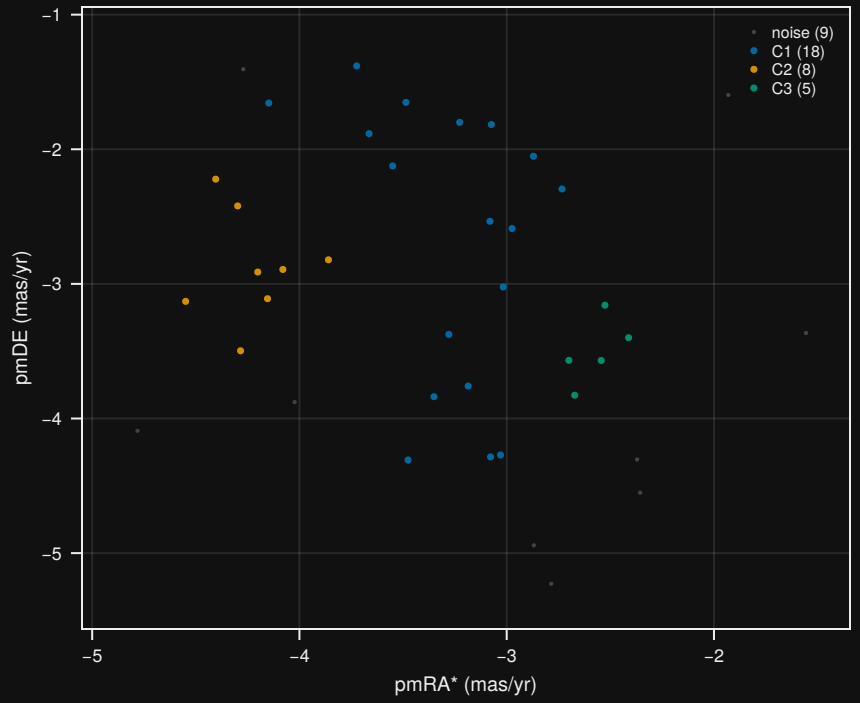


Parallax quality (n = 29)

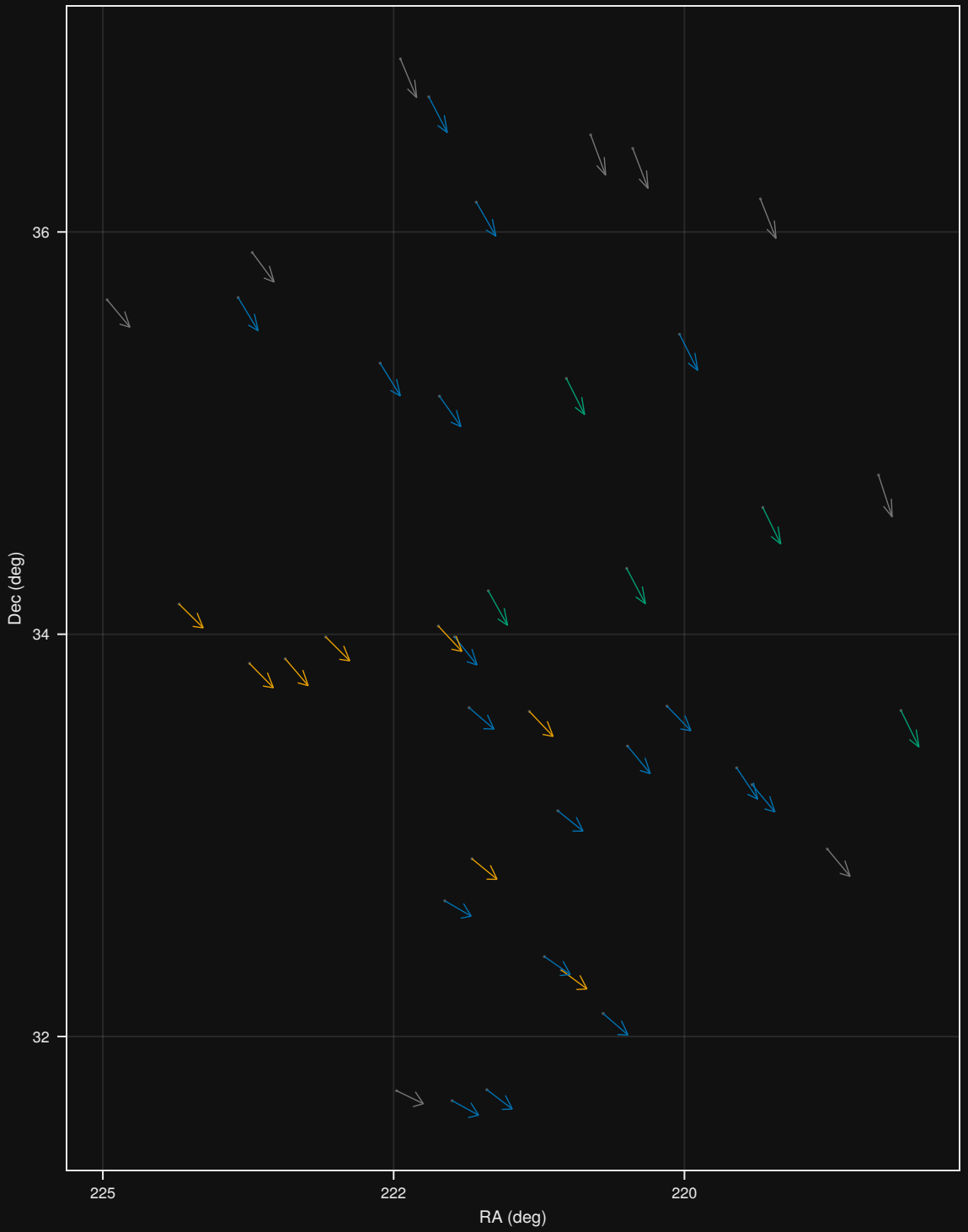


New-19

PM clustering · HDBSCAN · 3 cluster(s) · noise 22%



Proper-motion directions (arrows) coloured by cluster



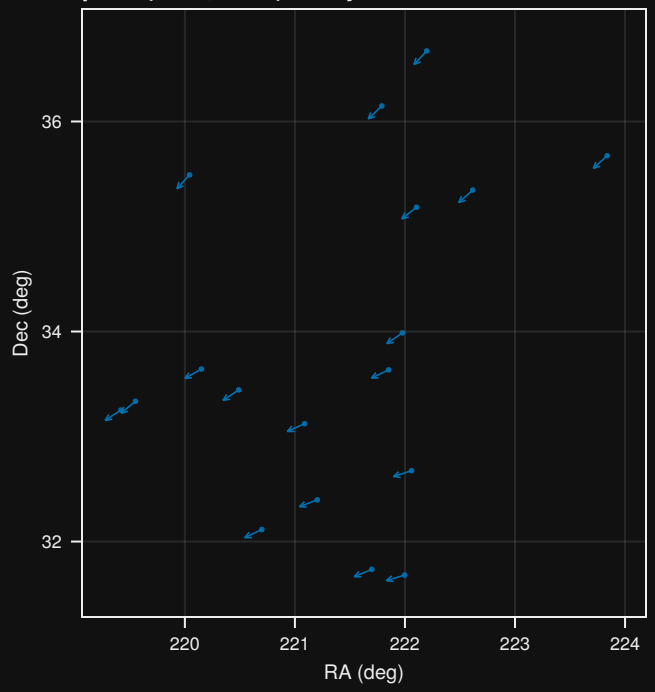
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	9	-2.79	-4.09	1.27	-0.80	2
C1	18	-3.21	-2.42	0.31	-0.64	1
C2	8	-4.24	-2.90	0.19	-0.77	0
C3	5	-2.54	-3.57	0.19	-0.54	1

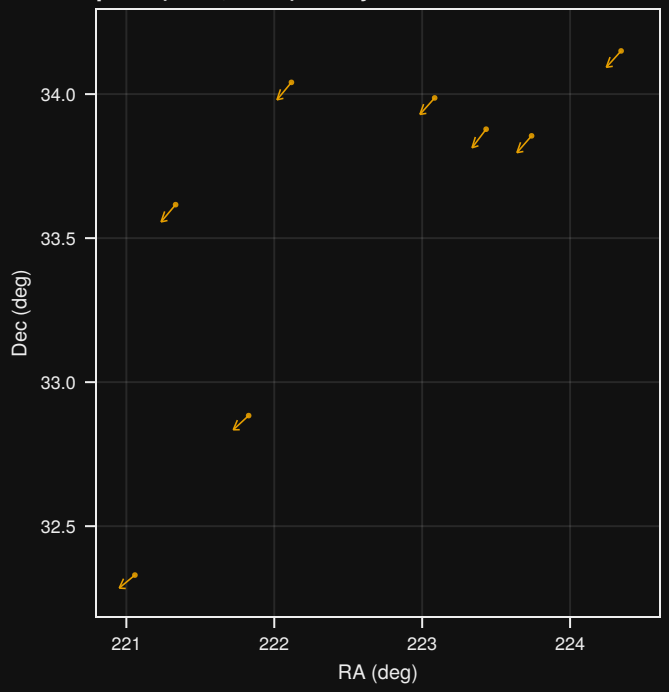
New-19

Per-cluster proper-motion vector maps

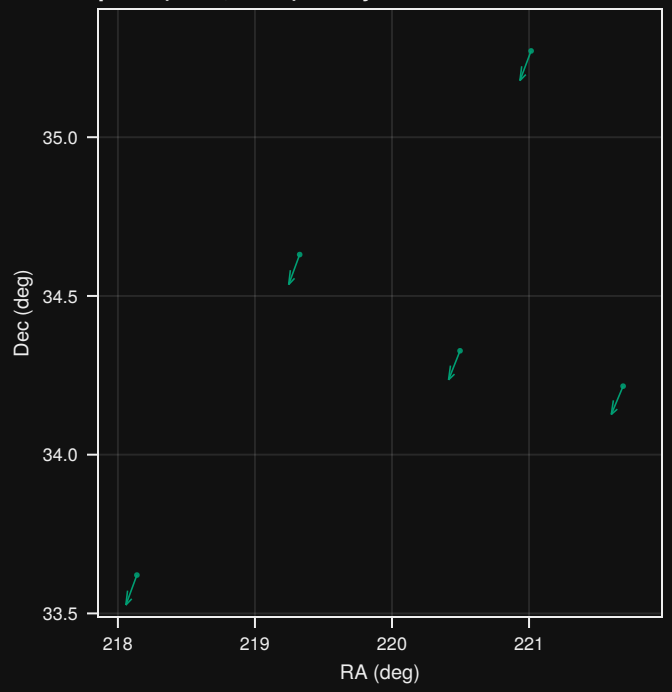
Cluster C1 · n = 18
pm = (-3.21, -2.42) mas/yr



Cluster C2 · n = 8
pm = (-4.24, -2.90) mas/yr



Cluster C3 · n = 5
pm = (-2.54, -3.57) mas/yr



Noise · n = 9

